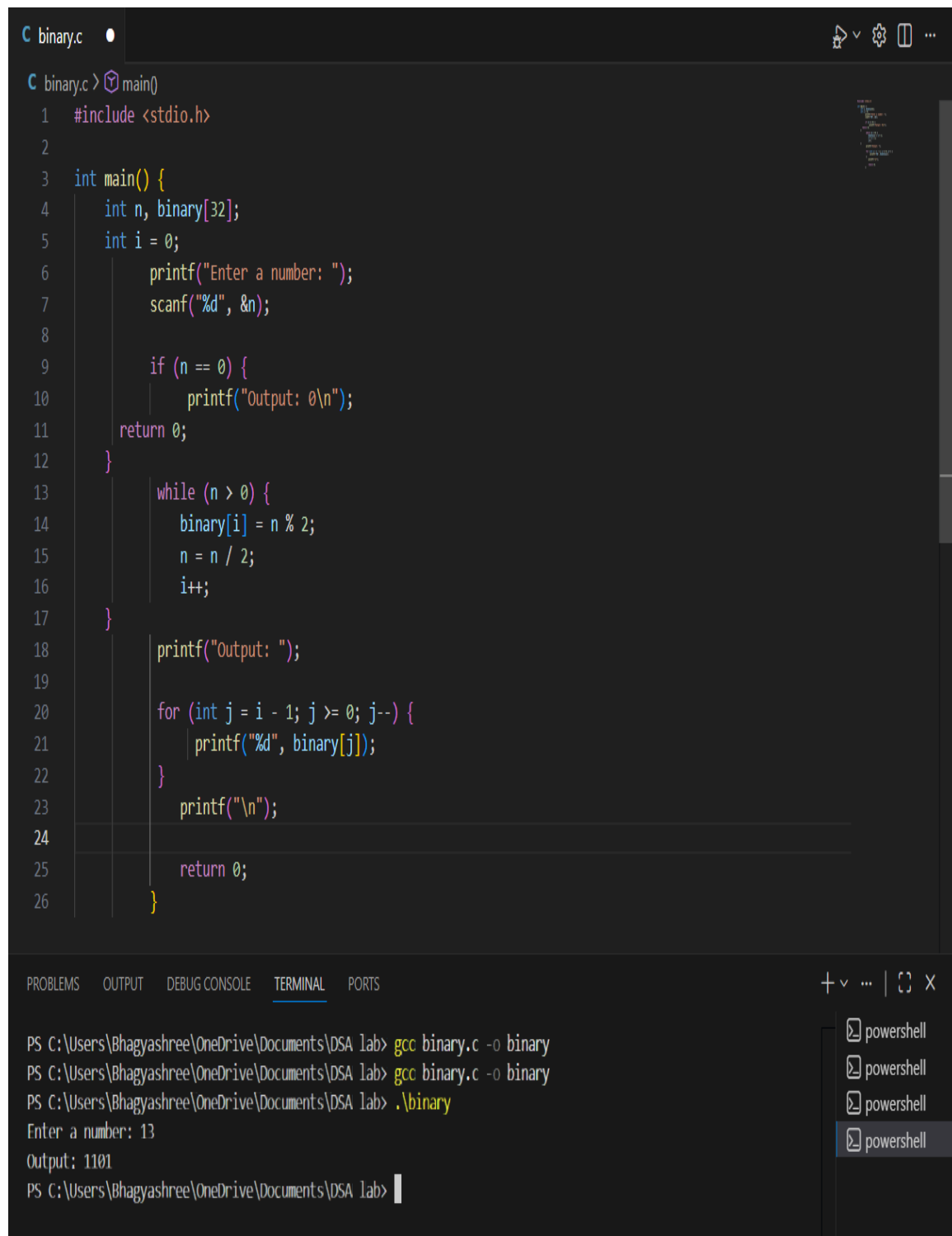


1. Write a program to print the binary representation of a number using loops.

Example: Input = 5, Output = 101.

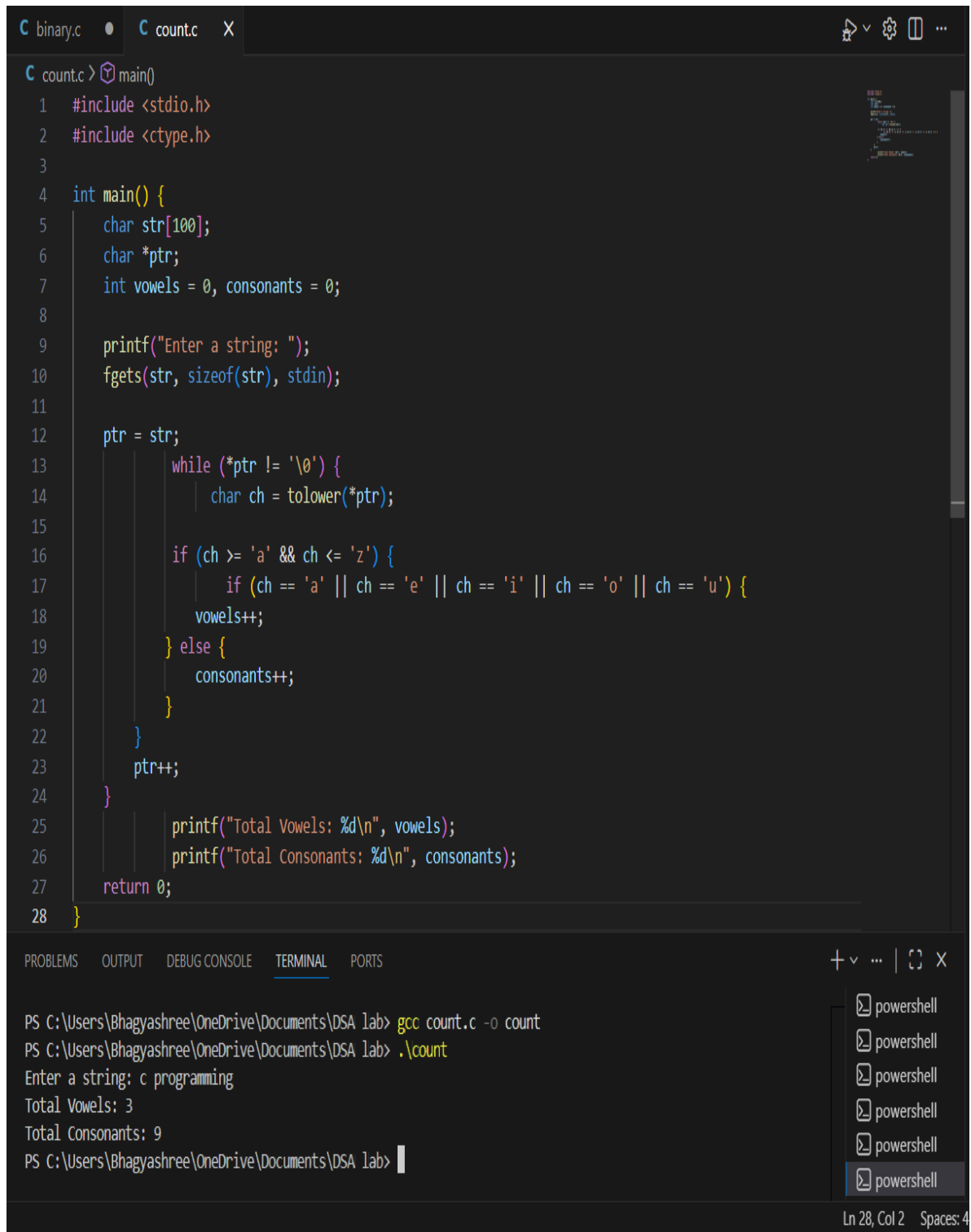


```
C binary.c
C binary.c > main()
1  #include <stdio.h>
2
3  int main() {
4      int n, binary[32];
5      int i = 0;
6      printf("Enter a number: ");
7      scanf("%d", &n);
8
9      if (n == 0) {
10         printf("Output: 0\n");
11         return 0;
12     }
13     while (n > 0) {
14         binary[i] = n % 2;
15         n = n / 2;
16         i++;
17     }
18     printf("Output: ");
19     for (int j = i - 1; j >= 0; j--) {
20         printf("%d", binary[j]);
21     }
22     printf("\n");
23
24
25     return 0;
26 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc binary.c -o binary
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc binary.c -o binary
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> .\binary
Enter a number: 13
Output: 1101
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab>
```

2. Write a program to count the number of vowels and consonants in a string using Pointer.



```
C binary.c ● C count.c X
C count.c > main()
1  #include <stdio.h>
2  #include <ctype.h>
3
4  int main() {
5      char str[100];
6      char *ptr;
7      int vowels = 0, consonants = 0;
8
9      printf("Enter a string: ");
10     fgets(str, sizeof(str), stdin);
11
12     ptr = str;
13     while (*ptr != '\0') {
14         char ch = tolower(*ptr);
15
16         if (ch >= 'a' && ch <= 'z') {
17             if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
18                 vowels++;
19             } else {
20                 consonants++;
21             }
22         }
23         ptr++;
24     }
25     printf("Total Vowels: %d\n", vowels);
26     printf("Total Consonants: %d\n", consonants);
27     return 0;
28 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc count.c -o count
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> ./count
Enter a string: c programming
Total Vowels: 3
Total Consonants: 9
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab>
```

Ln 28, Col 2 Spaces: 4

3. Write separate functions for addition, subtraction, multiplication, and division. Call these functions from a menu-driven program.

```
C binary.c  C count.c  C calculator.c X
C calculator.c > main()
1  #include <stdio.h>
2
3  float add(float a, float b) { return a + b; }
4  float sub(float a, float b) { return a - b; }
5  float mul(float a, float b) { return a * b; }
6  float div(float a, float b) {
7      return (b == 0) ? printf("Error: Div by 0!\n"), 0 : a / b;
8  }
9  int main() {
10     int choice;
11     float n1, n2;
12     do {
13         printf("\n1.Add 2.Sub 3.Mul 4.Div 5.Exit\nChoice: ");
14         scanf("%d", &choice);
15
16         if (choice >= 1 && choice <= 4) {
17             printf("Enter two numbers: ");
18             scanf("%f %f", &n1, &n2);
19         }
20         switch (choice) {
21             case 1: printf("Res: %.2f\n", add(n1, n2)); break;
22             case 2: printf("Res: %.2f\n", sub(n1, n2)); break;
23             case 3: printf("Res: %.2f\n", mul(n1, n2)); break;
24             case 4: printf("Res: %.2f\n", div(n1, n2)); break;
25             case 5: printf("Exiting...\n"); break;
26             default: printf("Invalid choice!\n");
27         }
28     } while (choice != 5);
29     return 0;
30 }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc calculator.c -o calculator
c:/mingw/bin/./lib/gcc/mingw32/6.3.0/./.././../mingw32/bin/ld.exe: cannot open output file calculator.exe: Permission denied
collect2.exe: error: ld returned 1 exit status
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> .\calculator

Simple Calculator
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 2
Enter two numbers: 21746
20244
Result: 1502.00
```

4. Write a program to print a diamond-shaped pattern using loops.

Ex: Input: n = 5

```
C binary.c C count.c C calculator.c C pattern.c X
C pattern.c > main()
1  #include <stdio.h>
2  int main() {
3      int n, i, j, space;
4
5      printf("Enter number of rows (for half diamond): ");
6      scanf("%d", &n);
7
8      for (i = 1; i <= n; i++) {
9          for (space = 1; space <= n - i; space++) {
10             printf(" ");
11         }
12         for (j = 1; j <= 2 * i - 1; j++) {
13             printf("*");
14         }
15         printf("\n");
16     }
17     for (i = n - 1; i >= 1; i--) {
18         for (space = 1; space <= n - i; space++) {
19             printf(" ");
20         }
21         for (j = 1; j <= 2 * i - 1; j++) {
22             printf("*");
23         }
24         printf("\n");
25     }
26     return 0;
27 }
```

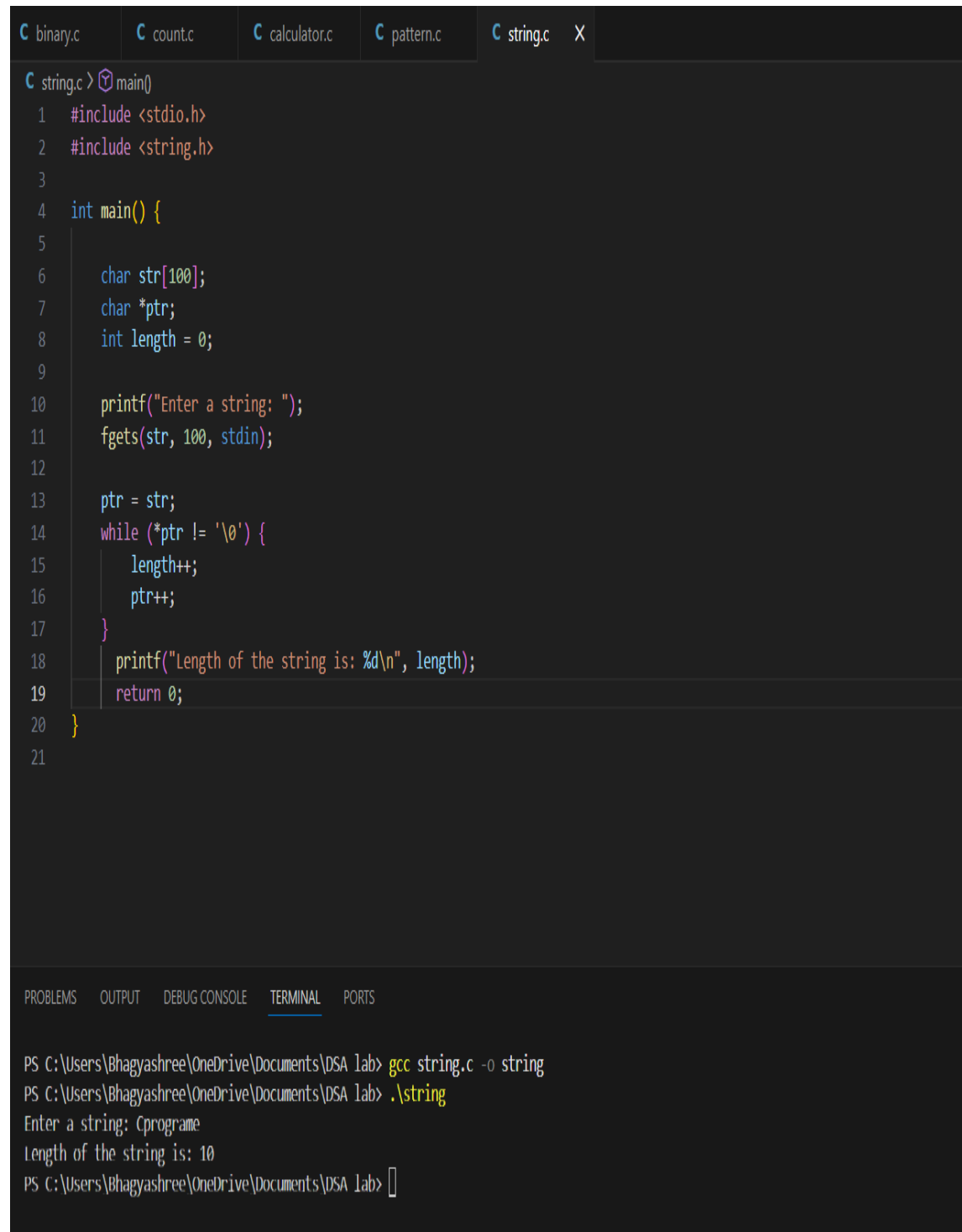
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> .\pattern
Enter number of rows (for half diamond): 3
*
***
*****
***
*
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab>
```

5. Write a program to find the length of a given string using pointers.

Ex : Input: Hello

Output: 5



```
C binary.c C count.c C calculator.c C pattern.c C string.c X
C string.c > main()
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5
6      char str[100];
7      char *ptr;
8      int length = 0;
9
10     printf("Enter a string: ");
11     fgets(str, 100, stdin);
12
13     ptr = str;
14     while (*ptr != '\0') {
15         length++;
16         ptr++;
17     }
18     printf("Length of the string is: %d\n", length);
19     return 0;
20 }
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

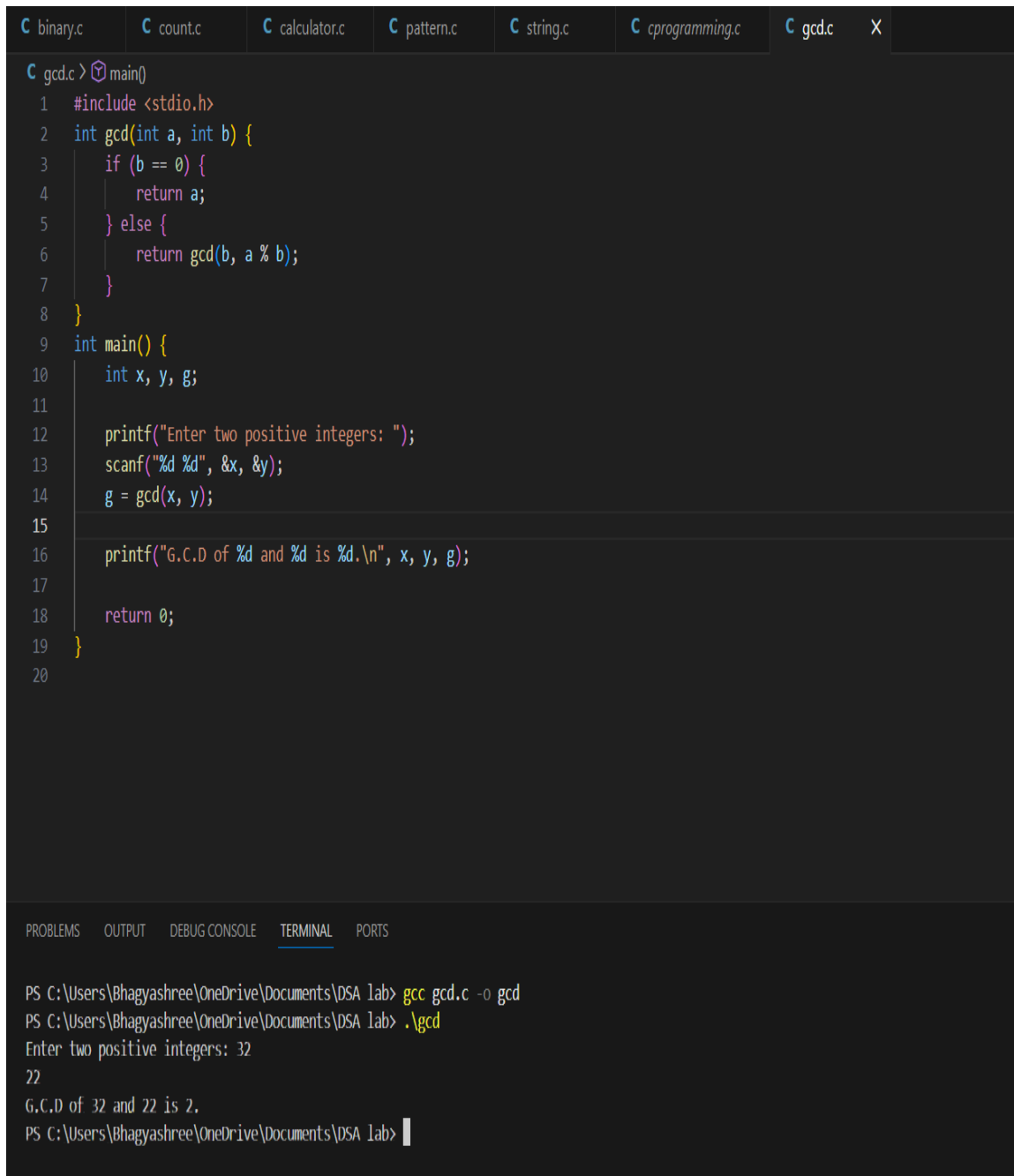
```
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc string.c -o string
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> .\string
Enter a string: Cprograme
Length of the string is: 10
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> 
```

6. Write a function to calculate the GCD (Greatest Common Divisor) of two numbers using

recursion.

Ex: Input: a = 56, b = 98

Output: GCD is 14.



The image shows a screenshot of a C++ IDE with a dark theme. At the top, there is a tab bar with several files: 'binary.c', 'count.c', 'calculator.c', 'pattern.c', 'string.c', 'cprogramming.c', and 'gcd.c'. The 'gcd.c' tab is active. Below the tab bar, the code for 'gcd.c' is displayed. It includes a recursive function 'gcd' and a 'main' function. The 'main' function prompts the user to enter two positive integers, reads them, and then prints the GCD. Below the code editor, there is a terminal window with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is selected, showing the command prompt output for compiling and running the program. The user has entered '32' and '22' as input, and the program has output 'G.C.D of 32 and 22 is 2.'

```
gcd.c > main()
1  #include <stdio.h>
2  int gcd(int a, int b) {
3      if (b == 0) {
4          return a;
5      } else {
6          return gcd(b, a % b);
7      }
8  }
9  int main() {
10     int x, y, g;
11
12     printf("Enter two positive integers: ");
13     scanf("%d %d", &x, &y);
14     g = gcd(x, y);
15
16     printf("G.C.D of %d and %d is %d.\n", x, y, g);
17
18     return 0;
19 }
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> gcc gcd.c -o gcd
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab> .\gcd
Enter two positive integers: 32
22
G.C.D of 32 and 22 is 2.
PS C:\Users\Bhagyashree\OneDrive\Documents\DSA lab>
```